

Disproof of conjecture 00000000086:  
 $\log |B(m, p)| \not\sim \frac{(m-1)^2}{2} \log p$ , already for  $m = 2$

Jackmeson1

2026-10-05

## 1 The conjecture and the reading

**Statement (English, verbatim).** Definition: The free Burnside group  $B(m, p)$  denotes the largest  $m$ -generated group satisfying  $x^p = 1$ . Conjecture:  $\log |B(m, p)|$  is asymptotic to  $(m - 1)^2/2 \cdot \log p$ ; and this asymptotic holds as  $p$  tends to infinity. (The Chinese text states the same.)

**Reading.** For fixed  $m$ , as  $p \rightarrow \infty$ ,

$$\log |B(m, p)| \sim \frac{(m-1)^2}{2} \log p,$$

where  $u \sim v$  means  $u - v = o(v)$  (Mathlib `Asymptotics.IsEquivalent`). The rank  $m$  is a free parameter, so the conjecture asserts this for every  $m$ ; we take the weakest such form, every  $m \geq 2$ , and refute it at  $m = 2$ . That refutes the universal claim for any range of  $m$  containing 2. We also refute the individual claims for  $m = 1$  and  $m = 3$ .

**Conventions.**  $\log$  is the natural logarithm. The base does not matter: changing it multiplies both sides by the same positive constant, and  $\sim$  is invariant under this. The exponent  $p$  ranges over the natural numbers. “ $p \rightarrow \infty$ ” is the filter `atTop`. We also treat  $p \rightarrow \infty$  through the primes, and in fact any filter  $l \leq \text{atTop}$  with  $l \neq \perp$ .  $|G|$  is `Nat.card G`, which is 0 when  $G$  is infinite (Mathlib’s junk value), so  $\log |G| = \log 0 = 0$  (Mathlib’s `Real.log 0 = 0`). We refute two readings of  $\log |B|$  for possibly infinite  $B$ :

- *Strict reading (Claim):*  $B(m, p)$  is finite for all large  $p$  (so that  $\log |B(m, p)|$  is a real number), and the asymptotic holds.
- *Junk reading:* the asymptotic holds for  $\log(\text{Nat.card})$ , with no finiteness requirement.

A reading in which  $\log |B| = +\infty$  for infinite  $B$  fails at every such  $p$  in the same way as the strict reading. We do not refute the claim for an individual  $m \geq 4$  (see Remark 10).

## 2 The free Burnside group (as formalized)

**Definition 1** (`B, gen`).  $B(m, p) = F_m/N$ , where  $F_m$  is the free group on  $\{0, \dots, m-1\}$  (`FreeGroup (Fin m)`) and  $N$  is the normal closure of the set  $\{g^p : g \in F_m\}$  (`powers m p`). The generators are the images  $x_i$  of the free generators.

**Definition 2** (`IsLargest m p G`). A group  $G$  is a *largest  $m$ -generated group satisfying  $x^p = 1$*  if (i)  $G$  is generated by some  $m$  elements, (ii)  $x^p = 1$  for every  $x \in G$ , and (iii) for every group  $H$  that satisfies (i) and (ii), there is a surjective homomorphism  $G \rightarrow H$ .

Wikipedia's article on the Burnside problem defines  $B(m, n)$  by this universal property: given any group  $G$  with  $m$  generators and of exponent  $n$ , "there is a unique homomorphism from  $B(m, n)$  to  $G$  that maps the  $i$ th generator  $x_i$  of  $B(m, n)$  into the  $i$ th generator  $g_i$  of  $G$ ". We prove exactly this property for our  $B(m, p)$ :

- Lemma 3.**    1.  $x^p = 1$  for every  $x \in B(m, p)$  (*B\_pow\_eq\_one*).  
 2.  $B(m, p)$  is generated by  $x_0, \dots, x_{m-1}$  (*closure\_range\_gen*).  
 3. For every group  $H$  with  $h^p = 1$  for all  $h \in H$ , and every map  $f$  from  $\{0, \dots, m-1\}$  to  $H$ , there is a homomorphism  $\text{lift } f : B(m, p) \rightarrow H$  with  $x_i \mapsto f(i)$  (*lift\_lift\_gen*). It is unique: two homomorphisms that agree on the  $x_i$  are equal (*hom\_ext*). If the  $f(i)$  generate  $H$ , then  $\text{lift } f$  is surjective (*lift\_surjective*).  
 4. Hence  $B(m, p)$  satisfies *IsLargest m p* (*B\_isLargest*).

*Proof.* (1) Every  $x$  is the image of some  $g \in F_m$ , and  $g^p \in N$ . (2) The free generators generate  $F_m$ , and the quotient map is surjective. (3)  $N$  lies in the kernel of the free-group lift of  $f$ , because  $f$ -images of  $p$ -th powers are  $p$ -th powers in  $H$ , hence trivial. Uniqueness holds because a homomorphism out of  $F_m/N$  is determined on  $F_m$ , and on  $F_m$  by the free generators. Surjectivity: the range is a subgroup that contains every  $f(i)$ .  $\square$

All results below are proved for an arbitrary family of groups satisfying *IsLargest*, so they do not depend on our particular construction. Lemma 3 shows that this hypothesis is satisfied.

### 3 The proof

**Lemma 4** (*E\_pow\_eq\_one, closure\_range\_egen, card\_E*). Let  $E(m, p) = (\mathbb{Z}/p)^m$ , written multiplicatively (*Multiplicative (Fin m -> ZMod p)*). Then  $x^p = 1$  for every  $x$ . For  $p \geq 1$ ,  $E(m, p)$  is generated by the  $m$  standard basis vectors. Also  $|E(m, p)| = p^m$ .

*Proof.*  $p \cdot a = 0$  in  $\mathbb{Z}/p$ . For the second claim,  $x = \sum_i x_i e_i$ , a product of powers of the basis vectors.  $\square$

**Lemma 5** (*card\_ge, finite\_of\_isLargest\_one*). Let  $G$  satisfy *IsLargest m p*, with  $p \geq 1$ . If  $G$  is finite, then  $|G| \geq p^m$ . If  $m = 1$ , then  $G$  is finite.

*Proof.* By (iii),  $G$  maps onto  $E(m, p)$ , so  $|G| \geq |E(m, p)| = p^m$ . If  $m = 1$ , then  $G$  is generated by one element  $g$  with  $g^p = 1$ , so  $G = \langle g \rangle$  is finite.  $\square$

**Lemma 6** (*not\_isEquivalent\_of\_gap*). Let  $l$  be a filter on  $\mathbb{N}$  with  $l \neq \perp$ , and let  $\eta > 0$ . If  $\eta|v(p)| < |u(p) - v(p)|$  for  $l$ -eventually all  $p$ , then  $u \not\sim v$  along  $l$ .

*Proof.*  $u \sim v$  would give  $|u(p) - v(p)| \leq \eta|v(p)|$  eventually. Both eventualities hold at some  $p$ , which is a contradiction.  $\square$

**Lemma 7** (*gap*). Let  $m \in \{1, 2, 3\}$ ,  $p \geq 2$ ,  $G$  satisfy *IsLargest m p*,  $L = \log p > 0$ ,  $\kappa = (m-1)^2/2 \in \{0, \frac{1}{2}, 2\}$  and  $v = \kappa L$ . Then  $\frac{1}{4}|v| < |\log |G| - v|$  (with  $\log |G| = 0$  if  $G$  is infinite).

*Proof.* If  $G$  is finite, Lemma 5 gives  $\log |G| \geq mL$ , so  $\log |G| - v \geq (m - \kappa)L$ . Here  $m - \kappa \in \{1, \frac{3}{2}, 1\}$ , which is  $> \frac{1}{4}\kappa \in \{0, \frac{1}{8}, \frac{1}{2}\}$ . If  $G$  is infinite, then  $m \in \{2, 3\}$  by Lemma 5,  $\log |G| = 0$ , and  $|0 - v| = \kappa L > \frac{1}{4}\kappa L$  since  $\kappa > 0$ .  $\square$

**Theorem 8** (`not_asymptotic_of_isLargest`, `burnside_not_asymptotic`, `burnside_not_claim`). Let  $l \leq \text{atTop}$  be a filter on  $\mathbb{N}$  with  $l \neq \perp$ , and let  $m \in \{1, 2, 3\}$ . For every family  $(G_p)$  of groups with  $\text{isLargest } m \text{ } p \text{ } G_p$  for all  $p$ , and in particular for  $G_p = B(m, p)$ , the function  $\log |G_p|$  (junk value 0 for infinite  $G_p$ ) is not asymptotic to  $\frac{(m-1)^2}{2} \log p$  along  $l$ . Consequently the strict reading *Claim*  $l \text{ } m \text{ } (B \text{ } m)$  is false.

*Proof.*  $p \geq 2$  holds  $l$ -eventually, since  $l \leq \text{atTop}$ . Apply the gap lemma with  $\eta = \frac{1}{4}$  and Lemma 7. The strict reading contains the asymptotic as a conjunct.  $\square$

**Theorem 9** (`conjecture86_false`, `conjecture86_false_atTop`, `conjecture86_false_primes`). *Conjecture*  $l$ , defined as “for every  $m \geq 2$ :  $B(m, p)$  is eventually finite along  $l$  and  $\log |B(m, p)| \sim \frac{(m-1)^2}{2} \log p$  along  $l$ ”, is false for every filter  $l \leq \text{atTop}$  with  $l \neq \perp$ . This includes  $l = \text{atTop}$  (all integers  $p \rightarrow \infty$ ) and  $l = \text{atTop} \sqcap \mathcal{P}\{p \text{ prime}\}$  (primes  $p \rightarrow \infty$ ). The filter of primes is not  $\perp$ , because there are infinitely many primes (`primes_neBot`).

*Proof.* Take  $m = 2$ . Concretely, if  $B(2, p)$  is finite then  $|B(2, p)| \geq p^2$ , so  $\log |B(2, p)| / (\frac{1}{2} \log p) \geq 4$  for every  $p \geq 2$ . The ratio cannot tend to 1.  $\square$

*Remark 10.* For  $m \geq 4$  we have  $m \leq (m-1)^2/2$ , and the abelian quotient no longer separates the two sides. We do not formalize the  $m \geq 4$  instances. As context only (not used in the proof): the Wikipedia article on the Burnside problem states that Ivanov’s theorem, together with the Novikov–Adian theorem, implies infiniteness for all  $m > 1$  and  $n \geq 2^{48}$  (the exponent is a superscript in the article; plain-text extraction renders it as “248”). Under the strict reading, this would refute every instance  $m \geq 2$ .

## 4 Lean formalization and axiom audit

The file `lean/Conjecture86/Basic.lean` (235 lines, Mathlib v4.33.1, only `import Mathlib`) contains all the definitions and theorems named above. The main statement is:

```
theorem conjecture86_false (l : Filter Nat) [l.NeBot]
  (hl : l <= Filter.atTop) : Not (Conjecture l)
def Conjecture (l : Filter Nat) : Prop :=
  forall m : Nat, 2 <= m -> Claim l m (B m)
def Claim (l : Filter Nat) (m : Nat) (G : Nat -> Type) : Prop :=
  Filter.Eventually (fun p => Finite (G p)) l /\
  Asymptotics.IsEquivalent l (logOrder G) (target m)
```

This is an ASCII rendering of the Lean source, which uses Unicode notation. Here `target m p = ((m : Real) - 1)^2 / 2 * Real.log p` and `logOrder G p = Real.log (Nat.card (G p))`. `lake build` succeeds with no `sorry` and no `native_decide`. `#print axioms` gives `[propext, Classical.choice, Quot.sound]` for each of `conjecture86_false`, `conjecture86_false_atTop`, `conjecture86_false_primes`, `burnside_not_asymptotic`, `not_asymptotic_of_isLargest`, `B_isLargest` and `hom_ext`.

## Source

Wikipedia, “Burnside problem”, [https://en.wikipedia.org/wiki/Burnside\\_problem](https://en.wikipedia.org/wiki/Burnside_problem) (retrieved 2026-10-05). It is used for the definition of  $B(m, n)$  and for the context in Remark 10.